Student's Name: **.....**

THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGION TERMINAL EXAMINATIONS - MAY, 2025
FORM THREE

SUBJECT CODE: 032 **CHEMISTRY** **Time: 3:00 Hours**

INSTRUCTIONS:

- This paper consists of sections A, B and C with a total of fourteen (14) questions
- Answer all questions in section A, B and C
- The following constants may be used where necessary
  - Atomic masses
    H = 1, C = 12, Na = 23, O = 16, Cl = 35.5, Al = 27, S = 32
  - Avogadro's constant = 6.02×10^{23}
  - GMV at s.t.p = $22.4dm^3$
- Non programmable calculators may be used

SECTION A: (16 MARKS) Answer **ALL** questions in this section

1. For each of the following items (i) – (x) choose the correct answer from the given alternatives and write its letter besides the item number in the answer sheet provided

   i) The three components of the fire triangle are
   A. Fuel, heat and carbon B. Fuel, Air and heat C.. Fuel, hydrogen and heat
   D. Fuel, nitrogen and heat E. fuel, oxygen and heat

   ii) Which of the following component is not likely to contain hydrogen
   A. Ammonia B. Water gas C. Producer gas D. Water E. Copper oxide

   iii) The maximum number of electrons in the inner most shell of an atom is
   A. 3 B. 2 C. 10 D. 8 E. 18

   iv) Zinc, lead and aluminum oxides are:-
   A. Acidic B. Basic C. Neutral d. Amphoteric E. Mixed oxide

   v) The tendency of a salt to lose its water of crystallization to the atmosphere is called:-
   A. Hydroscope B. Deliquescence C. Haber process D. Efflorescence E. Allotropy

   vi) An element X is found in period 4 and group II of the periodic table. If element X undergoes the reaction X – P x^{2+} +m$2^{e-}$the electronic configuration of x^{2+} ion is:-
   A. 2:8:8:2 B. 2:8:8 C. 2:8:6 D. 2:8:2 E. 2:8

   vii) A certain liquid has a PH of 3. this shows that it is an
   A. Alkali B. Acid C. Ion D. Oxide E. Neutral

   viii) A substance that can be easily compressed in its container
   A. Stone B. Water C. Oxygen gas D. Grease E. Nitrogen atom

   ix) The Brownian movement is taken to be an evidence of the:-
   A. Newton's theory B) Theory of colloidal suspension C) Kinetic theory of behavior of substance D) Theory of association of water molecules E) Theory of ionization

   x) A mixture of ammonium chloride and sand can be separated by using a method known as:-
   A. Evaporation B. Sorting C. Fractional distillation D. Sublimation E. Decantation

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2. Match the item in List A with the response in List B by writing the letter of the correct response beside the item number

LIST A	LIST B
(i) Method of separating mixture of a liquid and solid by pouring out the liquid and solid by pouring out the liquid only and the solid as a residue in a container (ii) Method of separating mixture of a solid and liquid by using a porous material (iii) Method of separating mixture using moving solvent material absorbs the solvent (iv) Method of separating mixture essential oil from plant materials (v) Method of separating soft water from cooking oil (vi) Method of separating two liquids with different boiling points	A. Distillation B. Chromatography C. Layer separation D. Filtration E. Sublimation F. Solvent extraction G. Evaporation H. Fractional distillation I. Decantation J. Magnetization K. Simple distillation

SECTION B: (54 MARKS) Answer all questions from this section

3. a) (i) Write the names and chemical formulas of the two chemicals that can be used in the preparation of oxygen gas.
   ii) Write down the balanced equation for the preparation of oxygen gas by using the chemical substance you have named in 3(a) (i) above
   iii) State the appropriate method of collecting oxygen gas based on the solubility of the gas in water
   b) i) With the aid of a chemical reaction explain why nitric acid is not used in the preparation of hydrogen gas in the laboratory.
   ii) Explain what will happen if a burning wooden splint is lowered in a test tube containing hydrogen gas
4. a) Differentiate between ionic and molecular equation
   b) Study the chemical equation below and answer the questions that follow

$$3KOH_{(aq)} + FeCl_{3(s)} \longrightarrow 3KCl_{(aq)} + Fe(OH)_{3(s)}$$

   i) Write the total ionic equation of this chemical equation
   ii) Write the net ionic equation for this chemical equation
   iii) What type of reaction is represented by the above equation
5. a) Explain why chemical and electrical conductivity of elements in the period table decreases across the periods
   b) Explain why cations slightly smaller than their neutral atoms, and anions are slightly larger than their neutral atoms
   c) Explain why water is said to be a universal solvent?
6. a) i) What is a standard solution?
   ii) Find the molarity of the sample of sulphuric acid containing 98% weight and having the specific gravity of $1.63g/cm^3$
   b) A solution was prepared by dissolving 2.62g of anhydrous sodium carbonate in water to make one litre of solution. It was observed that $25cm^3$ of this solution reacted with $23cm^3$ of the monobasic acid of the formula HQ containing 1.98g of the acid per litre of solution. Showing all the details of your calculations, find the relative atomic mass of the element Q
7. For each of the following reactions, write down a balanced chemical equation that represents each case
   a) The reaction between calcium carbonate and dilute hydrochloric acid
   b) Neutralization reaction between potassium hydroxide and dilute nitric acid
   c) Mixing Sodium chloride and Silver nitrate
   d) Combination reaction between nitrogen gas and hydrogen to form ammonia gas
   e) Displacement reaction between Zinc metal and dilute sulphuric acid
8. a) By using a balanced chemical equation, show how can you soften the following hard water?

i) Permanent hard water by using washing soda

ii) Temporary hard water by boiling

b) Briefly explain the following observation

i) There is a formation of white precipitates when carbon dioxide gas is bubbled through a bottle containing lime water

ii) When passed through ion exchanger became soft water though boiling didn't change it.

SECTION C: (30MARKS) Answer all question from this section

9. Chemistry as other science subject involves systematic ways of discovery of knowledge and solving of problems. Discovery of knowledge and solving problems involves very careful investigation. explain the validity of this statement

10. a) Define i) Mole ii) Molar solution

b) i) Calculate the amount of substance present in 11.2dm^3 of CO_2 at s.t.p

ii) Calculate the number of ions in 20g of aluminium sulphate (Al_2SO_4)

c) Find the volume of carbon dioxide produced when 5.6g of calcium carbonate is completely decomposed by heating at s.t.p.

11.Form Three students were conducting a practical in the laboratory, The practical involves reagents A and B,after the reactions, the results were universal solvent and rock salt only. From the practical conducted, identify the reaction and explain in details about five applications of identified reaction in our daily life